「Def. Let X_i $(i \in I)$ be Topological spaces with Topology $\mathcal{T}_i$, respectively.
Define a product subbasis on $\prod_{i\in I} X_i$ as:

$$S_p := \left\{ P_i^{-1}[U_i] \;\middle|\; i \in I,\ U_i \in \mathcal{T}_i \right\}$$

The Topology $\mathcal{T}_p$ generated by S_p is called a Product Topology on $\prod_{i\in I} X_i$

Clearly, the product Topology is the smallest Topology s.t. all Projections are Continuous. Moreover, on the Product Space, all Projections are Continuous open onto map.

「Thm. Let X, Y_i $(i \in I)$ be Topological Spaces. Define a map

$$f: X \to \prod_{i\in I} Y_i \;;\; x \mapsto (f_i(x))_{i\in I} \quad \text{where } f_i: X \to Y_i$$

Then, f is continuous if and only if for all $i \in I$, f_i is Continuous.

Proof If f is continuous, then $f_i = P_i \circ f$ is continuous, being composition of Continuous.
Conversely, let $B \in \beta_p$ be an base element of $\mathcal{T}_p$. Then, by definition,

$$B = \bigcap_{i=1}^{n} P_i^{-1}[U_i] \quad \text{for some } U_i \in \mathcal{T}_i,\ i = 1, \cdots, n.$$

Now,

$$f^{-1}[B] = f^{-1}\left[\bigcap_{i=1}^{n} P_i^{-1}[U_i]\right] = \bigcap_{i=1}^{n} \left(f^{-1} \circ P_i^{-1}\right)[U_i] = \bigcap_{i=1}^{n} (P_i \circ f)^{-1}[U_i] = \bigcap_{i=1}^{n} f_i[U_i]$$

□

Projection

[Embbeding

[Lemma.

Let X, Y be Topological Spaces, and let $y_0 \in Y$ be fixed. Define an Embbeding

$$E_x : X \to X \times Y : x \mapsto (x, y_0)$$

Then the E_x is an Embedding.

Proof. Let $U \subseteq X$ and $V \subseteq Y$ be open sets. Then, $U \times V \subseteq X \times Y$ is open set of the Product space. Then, the preimage of E_{y_0} is:

$$E_x^{-1}[U \times V] = \begin{cases} U & \text{if } y_0 \in V \\ \emptyset & \text{if } y_0 \notin V \end{cases}$$

Either case, the preimage is open set in X.

And, given an open set $U \subseteq X$, the image

$$E_x[U] = U \times \{y_0\} = (U \times Y) \cap (X \times \{y_0\})$$

is open set of $X \times \{y_0\}$. Finally, the injectivity is clear, so proved. □

[Thm.

Let $f : X \times Y \to Z$ be a continuous map. Then, for fixed $y_0 \in Y$, the map

$$f_x : X \to Z : x \mapsto f(x, y_0)$$

is continuous.

Proof. Since $f_x = f \circ E_x$ where E_x is the embbedings for fixed $y_0 \in Y$, it is continuous, being the composition of continuous maps. ⊠

「Thm. Let X_i $(i\in I)$ be Topological spaces, $A_i \subset X_i$. Then,

1) $\overline{\prod_{i\in I} A_i} = \prod_{i\in I} \overline{A_i}$

2) $\left(\prod_{i\in I} A_i\right)^\circ \subseteq \prod_{i\in I} A_i^\circ$, the equality holds when the product is finite.

Pf. $(x_i)\in \overline{\prod_{i\in I} A_i}$ $\Leftrightarrow$ $\forall\, \mathcal{U}\in \beta_p$ with $(x_i)\in\mathcal{U}$, $\mathcal{U}\cap \prod_{i\in I} A_i \neq \emptyset$

$\Leftrightarrow$ $\forall\, \mathcal{U}\in\beta_p$ with $(x_i)\in\mathcal{U}$, $\prod_{i\in I} \pi_i[\mathcal{U}] \cap \prod_{i\in I} A_i = \prod_{i\in I} \pi_i[\mathcal{U}]\cap A_i \neq \emptyset$

$\Leftrightarrow$ $\forall i\in I$, $\forall\, \mathcal{U}_i\in\beta_i$ with $x_i\in\mathcal{U}_i$, $\mathcal{U}_i \cap A_i \neq \emptyset$

$\Leftrightarrow$ $(x_i)\in \prod_{i\in I} \overline{A_i}$

(I finite)

$(x_i)\in \left(\prod_{i\in I} A_i\right)^\circ \overset{(*)}{\Leftrightarrow} \exists\, \mathcal{U}\in\beta_p$ s.t. $(x_i)\in\mathcal{U}\subseteq \prod_{i\in I} A_i$

$\Leftrightarrow$ $\forall i\in I$, $x_i \in p_i[\mathcal{U}] \subseteq A_i$

$\Leftrightarrow$ $\forall i\in I$, $x_i\in A_i^\circ$.

But, if I is Not finite, $(*)$ is broke.